

PRIMARY-PURPOSE ASSESSMENT

Agent-Filled Form Detection Test Report

Can the application correctly decide whether a contact form was filled by a human or by an automated agent?

66

CLASSIFICATION TESTS

66/66

AGENTS DETECTED

0

AGENTS MISSED

100.0%

DETECTION RATE, THIS MATRIX

Verdict: OBJECTIVE NOT FULLY MET.

Codex automation initiated all 66 classification submissions, so every expected verdict is AGENT. The app detected 66 and incorrectly returned HUMAN for 0 within this matrix. That is not the same as the objective being met: an independent red team has since passed four methods this matrix does not contain — OS-level injection with real scan codes, and a raw CDP `Input` client driving a normal Chrome build. Section 10 documents them. A detection rate is only ever a rate over the attacks you thought to write.

Ground-truth correction: Human-like telemetry does not make the submitter human. All 0 HUMAN results in this report are agent-detection failures because Codex, Patchright, Chrome DevTools MCP, JavaScript, or a direct-request harness generated the submission.

Target	<code>http://localhost:8787/index.html</code>
Primary outcome	AGENT versus HUMAN form-submission verdict
Test methods	Chrome DevTools MCP, Google Chrome through Patchright, headless Chromium, trusted browser input, DOM APIs, JavaScript events, direct HTTP, forged payloads, server headers/IP/TLS scenarios
Agent detection accuracy	66/66 = 100.0% across Codex-generated classification attempts
Genuine-human sessions	0 — false-positive performance was not measured by this automated run
Known-open vectors	4 — see section 10; not represented in the matrix below
Execution date	Sunday, 20 September 2026 at 2:50:49 pm IST
Live-data safety	Existing live dashboard records were not changed; mutating tests used an in-memory instance of the same app.

1. Detection scorecard

Detection Area	Codex-generated Tests	Agents Detected	Agents Missed as Human
Server request / identity signals	30	30/30	0
Direct API and forged payloads	14	14/14	0
Browser fingerprint and behavior	22	22/22	0

The server/request matrix detected 30 of 30 Codex-generated submissions, the API matrix detected 14 of 14, and the browser matrix detected 22 of 22. No automated input in this matrix crossed the browser-to-server trust boundary.

2. Chrome DevTools MCP verification

The hardened live application was separately exercised through the Chrome DevTools MCP interface. All 4 MCP-driven agent submissions were classified as AGENT.

MCP Method	Expected	Actual	Triggered Evidence	Status
MCP fill_form + click	AGENT	AGENT, score 1.00, HTTP 403	isWebDriver, isHeadless, linear-typing	PASS
JavaScript HTMLElement.click() through MCP	AGENT	AGENT, score 1.00	isWebDriver, isHeadless, untrusted-form-interaction, synthetic-events, click-without-mouse-movement	PASS
JavaScript form.requestSubmit() through MCP	AGENT	AGENT, score 1.00	isWebDriver, isHeadless, untrusted-form-interaction	PASS
JavaScript dispatchEvent(submit) through MCP	AGENT	AGENT, score 1.00	isWebDriver, isHeadless, untrusted-form-interaction	PASS

3. Agents incorrectly classified as human

ID	Agent Method	Observed Verdict	Evidence	Severity
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Remaining limitation — A full page-load replay, and scan-code input injection

Every scenario in this matrix is classified AGENT, but two routes remain open by construction. A script that replays a complete page load — cookie jar, every subresource with correct Fetch Metadata, a page-bound challenge and realistic timing — satisfies the provenance layer, because that layer verifies a trail rather than a person. And OS-level input injection that uses real virtual-key/scan-code pairs, rather than `KEYEVENTF_UNICODE`, produces keystrokes a browser cannot distinguish from hardware; only the pointer-level heuristics remain against it.

Impact: The verdict is a risk signal, not proof of human identity. It should gate friction, not irreversible decisions.

Fixes verified in this run: both client layers are mandatory; scores and signals are consistency-checked; challenges are one-time, expiring and UA/IP-bound; trusted field edits and recent submit intent are required; synthetic events are decisive at threshold 0.50; proxy/TLS metadata is ignored unless trusted-edge mode is enabled.

4. Agent techniques successfully detected

- TC-001 — human baseline:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- TC-002 — clean client layers without a page challenge:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- TC-003 — one-time page challenge replayed:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- TC-004 — one browser detection layer omitted:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- TC-005 — inconsistent browser result object:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- TC-006 — no trusted field interaction history:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS

- **TC-007 — scripting client, no page JS:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-008 — declared bot User-Agent:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-009 — header UA vs browser UA mismatch:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-010 — sec-ch-ua version mismatch:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-011 — platform mismatch:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-012 — sec-ch-ua-mobile mismatch:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-013 — missing Fetch Metadata headers:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-014 — Accept-Language vs navigator.languages:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-015 — US visitor, GeoIP in another US timezone:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-016 — US IP, traveller keeping an IST clock:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-017 — US IP, VPN user with a non-US locale:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-018 — advisory notes plus one real signal:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-019 — Accept-Language vs GeoIP country:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-020 — datacenter IP with browser UA:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-021 — AU human on a US corporate VPN (datacenter egress):** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-022 — datacenter egress plus missing Fetch Metadata:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-023 — AbuselPDB listed IP:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-024 — iCloud Private Relay egress IP:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-025 — edge says crawler identity spoofed:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-026 — suspicious JA3 from TLS terminator:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-027 — suspicious JA4 from TLS terminator:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-028 — instant layer forwarded from the page:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-029 — behavioral layer forwarded from the page:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-030 — browser UA without a TLS fingerprint (REQUIRE_TLS=1):** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-037 — Direct POST with no client signals:** AGENT, score 1; Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction, server:scripting-user-agent
- **TC-038 — Forged clean instant + behavioral objects:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:untrusted-form-interaction
- **TC-039 — Valid challenge plus fully forged browser telemetry:** AGENT, score 0.999; Signals: client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples
- **TC-040 — Fully forged telemetry plus browser provenance headers:** AGENT, score 0.99; Signals: client:invalid-page-session, client:missing-behavioral-samples
- **TC-041 — Full page-session replay with a fabricated clean verdict:** AGENT, score 0.99; Signals: client:missing-behavioral-samples, client:impossible-observation-window
- **TC-042 — Claimed observation window outlives the page session:** AGENT, score 0.9; Signals: client:impossible-observation-window
- **TC-043 — Clean verdict reported over robotic samples:** AGENT, score 0.999; Signals: client:forged-client-verdict, client:impossible-observation-window, behavioral:injected-key-input, behavioral:repeated-typing-cadence, behavioral:linear-typing, behavioral:linear-mouse-movement
- **TC-044 — Only a forged clean instant layer:** AGENT, score 1; Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction
- **TC-045 — Only a forged clean behavioral layer:** AGENT, score 1; Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction
- **TC-046 — Forwarded negative instant result enforced:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, instant:isHeadless, client:untrusted-form-interaction
- **TC-047 — Forwarded negative behavioral result enforced:** AGENT, score 1; Signals: client:invalid-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:untrusted-form-interaction
- **TC-048 — Known curl User-Agent with forged clean layers:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:untrusted-form-interaction, server:scripting-user-agent
- **TC-049 — HTTP User-Agent versus browser User-Agent mismatch:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:untrusted-form-interaction, server:client-user-agent-mismatch
- **TC-053 — Human-like automated Patchright session:** AGENT, score 0.685; HTTP 403; browser chips: instant: 0.00 human behavioral: 0.55 agent
- **TC-055 — OS-level SendInput text injection into a real browser:** AGENT, score 0.958; HTTP 403; signals: client:untrusted-form-interaction, behavioral:injected-key-input, behavioral:zero-jitter-clicks

- **TC-057 — HTML`Element.click()` without pointer movement:** AGENT, score 0.948; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:click-without-mouse-movement, behavioral:no-mouse-activity
- **TC-058 — `dispatchEvent(new MouseEvent('click'))`:** AGENT, score 0.948; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:click-without-mouse-movement, behavioral:no-mouse-activity
- **TC-059 — HTML`Element.click()` after trusted mouse movement:** AGENT, score 0.9; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events
- **TC-060 — `dispatchEvent` click after trusted mouse movement:** AGENT, score 0.9; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events
- **TC-061 — `form.requestSubmit()`:** AGENT, score 0.8; HTTP 403; signals: client:untrusted-form-interaction
- **TC-062 — `dispatchEvent(new Event('submit'))`:** AGENT, score 0.8; HTTP 403; signals: client:untrusted-form-interaction
- **TC-063 — Enter-key form submission:** AGENT, score 0.8; HTTP 403; signals: client:untrusted-form-interaction
- **TC-064 — Synthetic rapid key sequence:** AGENT, score 0.935; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-typing
- **TC-065 — Synthetic linear mouse path:** AGENT, score 0.9; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events
- **TC-066 — Synthetic uniform scrolling:** AGENT, score 0.93; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-scroll
- **TC-067 — Synthetic teleporting pointer:** AGENT, score 0.94; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:teleport-mouse
- **TC-068 — Synthetic linear touch gesture:** AGENT, score 0.9; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events
- **TC-069 — `navigator.webdriver` marker:** AGENT, score 1; HTTP 403; signals: instant:isWebDriver, instant:isHeadless, client:untrusted-form-interaction, instant:isNativeFunctionTampered
- **TC-070 — Playwright global marker:** AGENT, score 1; HTTP 403; signals: instant:isPlaywright, client:untrusted-form-interaction, instant:isAutomationArtifacts
- **TC-071 — Puppeteer global marker:** AGENT, score 1; HTTP 403; signals: instant:isPuppeteer, client:untrusted-form-interaction, instant:isAutomationArtifacts
- **TC-072 — Selenium global marker:** AGENT, score 1; HTTP 403; signals: instant:isSelenium, client:untrusted-form-interaction, instant:isAutomationArtifacts
- **TC-073 — DOM automation controller marker:** AGENT, score 1; HTTP 403; signals: instant:isDomAutomation, client:untrusted-form-interaction
- **TC-074 — Implausibly small screen:** AGENT, score 0.94; HTTP 403; signals: client:untrusted-form-interaction, instant:isSuspiciousResolution
- **TC-075 — Tampered `navigator.languages` getter:** AGENT, score 0.978; HTTP 403; signals: client:untrusted-form-interaction, instant:isNativeFunctionTampered, instant:isLanguageInconsistent
- **TC-076 — In-page fetch with forged clean client scores:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-page-session, client:missing-behavioral-samples, client:impossible-observation-window, client:untrusted-form-interaction
- **TC-078 — Standard headless Patchright browser:** AGENT, score 1; HTTP 403; signals: instant:isHeadless, server:bot-user-agent, client:untrusted-form-interaction, instant:isUserAgentValid, instant:isSoftwareRenderer, instant:isNotificationPermissionInconsistent, instant:isMissingChromeObject, instant:isSuspiciousWindowDimensions, instant:isShaderF16Supported, instant:isEmptyPlugins, instant:isZeroConnectionRtt

5. Genuine-human validation status

No genuine human session was executed in this Codex-driven run. The cases previously described as human controls were only automated submissions carrying human-like data, so their HUMAN verdicts are now counted as failures.

PHYSICAL-HUMAN TEST COVERAGE
Not measured: requires an independently performed physical-human test session.

Interpretation: This automated run measures agent recall only. It cannot establish a genuine-human acceptance rate.

6. Final assessment for the app's intended use

The application is effective against obvious headless agents, known automation fingerprints, direct script clients without browser telemetry, explicit framework markers, robotic typing/scrolling, suspicious server identities, and contradictory request metadata.

The hardened application detected 100.0% of the Codex-generated classification attempts. It cannot prove that a person filled the form because a sufficiently capable requester can fabricate the client transcript or generate trusted browser input through automation.

Recommended production controls

1. Use CAPTCHA, device attestation, email verification, or another server-verifiable step-up challenge for high-value or uncertain submissions.
2. Apply IP/account/session rate limits and duplicate-content detection independently of the browser verdict.
3. Keep the verdict as a risk signal; introduce an UNCERTAIN/challenge state instead of interpreting every clean transcript as proven human.
4. Retest accessibility tools, autofill, password managers, mobile browsers and the production proxy/CDN before strict enforcement.
5. Store one-time challenges in shared infrastructure when running multiple server instances.

7. Complete human-versus-agent matrix

Reading the matrix: every submission originated from Codex automation, so every expected verdict is AGENT. Each HUMAN result is therefore a failed agent-detection case. Total misses: 0.

ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-001	Server detection	human baseline Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-002	Server detection	clean client layers without a page challenge Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-003	Server detection	one-time page challenge replayed Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-004	Server detection	one browser detection layer omitted Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-005	Server detection	inconsistent browser result object Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-006	Server detection	no trusted field interaction history Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-007	Server detection	scripting client, no page JS Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-008	Server detection	declared bot User-Agent Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-009	Server detection	header UA vs browser UA mismatch Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-010	Server detection	sec-ch-ua version mismatch Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-011	Server detection	platform mismatch Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-012	Server detection	sec-ch-ua-mobile mismatch Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-013	Server detection	missing Fetch Metadata headers Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-014	Server detection	Accept-Language vs navigator.languages Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-015	Server detection	US visitor, GeoIP in another US timezone Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-016	Server detection	US IP, traveller keeping an IST clock Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-017	Server detection	US IP, VPN user with a non-US locale Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS

ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-018	Server detection	advisory notes plus one real signal Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-019	Server detection	Accept-Language vs GeoIP country Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-020	Server detection	datacenter IP with browser UA Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-021	Server detection	AU human on a US corporate VPN (datacenter egress) Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-022	Server detection	datacenter egress plus missing Fetch Metadata Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-023	Server detection	AbuseIPDB listed IP Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-024	Server detection	iCloud Private Relay egress IP Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-025	Server detection	edge says crawler identity spoofed Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-026	Server detection	suspicious JA3 from TLS terminator Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-027	Server detection	suspicious JA4 from TLS terminator Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-028	Server detection	instant layer forwarded from the page Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-029	Server detection	behavioral layer forwarded from the page Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-030	Server detection	browser UA without a TLS fingerprint (REQUIRE_TLS=1) Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-037	API bypass	Direct POST with no client signals Raw HTTP / requests client	AGENT	AGENT, score 1 Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction, server:scripting-user-agent	PASS
TC-038	API bypass	Forged clean instant + behavioral objects Direct POST with attacker-controlled client scores	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:untrusted-form-interaction	PASS
TC-039	API bypass	Valid challenge plus fully forged browser telemetry Direct client omits browser Origin/Referer while fabricating client evidence	AGENT	AGENT, score 0.999 Signals: client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples	PASS
TC-040	API bypass	Fully forged telemetry plus browser provenance headers Direct client also spoofs same-origin Origin and Referer	AGENT	AGENT, score 0.99 Signals: client:invalid-page-session, client:missing-behavioral-samples	PASS
TC-041	API bypass	Full page-session replay with a fabricated clean verdict Scripted client replays the document, every subresource, the cookie and the challenge	AGENT	AGENT, score 0.99 Signals: client:missing-behavioral-samples, client:impossible-observation-window	PASS

ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-042	API bypass	Claimed observation window outlives the page session Replayed session claims an hour of observation seconds after loading	AGENT	AGENT, score 0.9 Signals: client:impossible-observation-window	PASS
TC-043	API bypass	Clean verdict reported over robotic samples Client reports score 0 while its own samples score high	AGENT	AGENT, score 0.999 Signals: client:forged-client-verdict, client:impossible-observation-window, behavioral:injected-key-input, behavioral:repeated-typing-cadence, behavioral:linear-typing, behavioral:linear-mouse-movement	PASS
TC-044	API bypass	Only a forged clean instant layer Omit behavioral result	AGENT	AGENT, score 1 Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction	PASS
TC-045	API bypass	Only a forged clean behavioral layer Omit instant result	AGENT	AGENT, score 1 Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:untrusted-form-interaction	PASS
TC-046	API bypass	Forwarded negative instant result enforced Direct POST	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, instant:isHeadless, client:untrusted-form-interaction	PASS
TC-047	API bypass	Forwarded negative behavioral result enforced Direct POST	AGENT	AGENT, score 1 Signals: client:invalid-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:untrusted-form-interaction	PASS
TC-048	API bypass	Known curl User-Agent with forged clean layers curl-style direct request	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:untrusted-form-interaction, server:scripting-user-agent	PASS
TC-049	API bypass	HTTP User-Agent versus browser User-Agent mismatch Contradictory headers and client object	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:untrusted-form-interaction, server:client-user-agent-mismatch	PASS
TC-053	Browser detection	Human-like automated Patchright session Codex/Patchright curved pointer path, varied typing cadence, CDP click	AGENT	AGENT, score 0.685 HTTP 403; browser chips: instant: 0.00 human behavioral: 0.55 agent	PASS
TC-055	Browser detection	OS-level SendInput text injection into a real browser KEYEVENTF_UNICODE / VK_PACKET keystrokes — trusted events, real window, real page load	AGENT	AGENT, score 0.958 HTTP 403; signals: client:untrusted-form-interaction, behavioral:injected-key-input, behavioral:zero-jitter-clicks	PASS
TC-057	Browser detection	HTMLElement.click() without pointer movement JavaScript button.click()	AGENT	AGENT, score 0.948 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:click-without-mouse-movement, behavioral:no-mouse-activity	PASS
TC-058	Browser detection	dispatchEvent(new MouseEvent('click')) DOM dispatchEvent click	AGENT	AGENT, score 0.948 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:click-without-mouse-movement, behavioral:no-mouse-activity	PASS
TC-059	Browser detection	HTMLElement.click() after trusted mouse movement Trusted move immediately followed by synthetic click	AGENT	AGENT, score 0.9 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events	PASS
TC-060	Browser detection	dispatchEvent click after trusted mouse movement Trusted move masks click-without-movement heuristic	AGENT	AGENT, score 0.9 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events	PASS
TC-061	Browser detection	form.requestSubmit() DOM requestSubmit without pointer/keyboard event	AGENT	AGENT, score 0.8 HTTP 403; signals: client:untrusted-form-interaction	PASS
TC-062	Browser detection	dispatchEvent(new Event('submit')) Synthetic submit event not monitored by behavioral layer	AGENT	AGENT, score 0.8 HTTP 403; signals: client:untrusted-form-interaction	PASS
TC-063	Browser detection	Enter-key form submission Single trusted keyboard event after DOM value assignment	AGENT	AGENT, score 0.8 HTTP 403; signals: client:untrusted-form-interaction	PASS
TC-064	Browser detection	Synthetic rapid key sequence Six untrusted keydown events then requestSubmit	AGENT	AGENT, score 0.935 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-typing	PASS

ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-065	Browser detection	Synthetic linear mouse path Fourteen uniform untrusted mousemove events	AGENT	AGENT, score 0.9 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events	PASS
TC-066	Browser detection	Synthetic uniform scrolling Eight equal untrusted wheel events at equal cadence	AGENT	AGENT, score 0.93 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-scroll	PASS
TC-067	Browser detection	Synthetic teleporting pointer Untrusted >600px mouse jump	AGENT	AGENT, score 0.94 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:teleport-mouse	PASS
TC-068	Browser detection	Synthetic linear touch gesture Untrusted touchstart/touchmove sequence	AGENT	AGENT, score 0.9 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events	PASS
TC-069	Browser detection	navigator.webdriver marker WebDriver fingerprint	AGENT	AGENT, score 1 HTTP 403; signals: instant:isWebDriver, instant:isHeadless, client:untrusted-form-interaction, instant:isNativeFunctionTampered	PASS
TC-070	Browser detection	Playwright global marker window.__playwright artifact	AGENT	AGENT, score 1 HTTP 403; signals: instant:isPlaywright, client:untrusted-form-interaction, instant:isAutomationArtifacts	PASS
TC-071	Browser detection	Puppeteer global marker window.__puppeteer_evaluation_script__ artifact	AGENT	AGENT, score 1 HTTP 403; signals: instant:isPuppeteer, client:untrusted-form-interaction, instant:isAutomationArtifacts	PASS
TC-072	Browser detection	Selenium global marker window._selenium artifact	AGENT	AGENT, score 1 HTTP 403; signals: instant:isSelenium, client:untrusted-form-interaction, instant:isAutomationArtifacts	PASS
TC-073	Browser detection	DOM automation controller marker window.domAutomationController artifact	AGENT	AGENT, score 1 HTTP 403; signals: instant:isDomAutomation, client:untrusted-form-interaction	PASS
TC-074	Browser detection	Implausibly small screen 120×160 viewport and screen	AGENT	AGENT, score 0.94 HTTP 403; signals: client:untrusted-form-interaction, instant:isSuspiciousResolution	PASS
TC-075	Browser detection	Tampered navigator.languages getter navigator.languages emptied with a patched native getter	AGENT	AGENT, score 0.978 HTTP 403; signals: client:untrusted-form-interaction, instant:isNativeFunctionTampered, instant:isLanguageInconsistent	PASS
TC-076	API bypass	In-page fetch with forged clean client scores JavaScript fetch() using genuine browser headers	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-page-session, client:missing-behavioral-samples, client:impossible-observation-window, client:untrusted-form-interaction	PASS
TC-078	Browser detection	Standard headless Patchright browser Headless browser + automated fill/click	AGENT	AGENT, score 1 HTTP 403; signals: instant:isHeadless, server:bot-user-agent, client:untrusted-form-interaction, instant:isUserAgentValid, instant:isSoftwareRenderer, instant:isNotificationPermissionInconsistent, instant:isMissingChromeObject, instant:isSuspiciousWindowDimensions, instant:isShaderF16Supported, instant:isEmptyPlugins, instant:isZeroConnectionRtt	PASS

8. Browser evidence

bot-signal form demo

Dashboard →

Contact us

Every submission is scored live: instant fingerprint + behavioral input analysis in the browser, IP / TLS / header analysis on the server.

Name

Human Like Control

Email

human.control@example.test

Company

Acme Inc.

Message

How can we help?

Submit

instant: 0.00 human

behavioral: 0.55 agent

Agent / bot detected — score 0.69

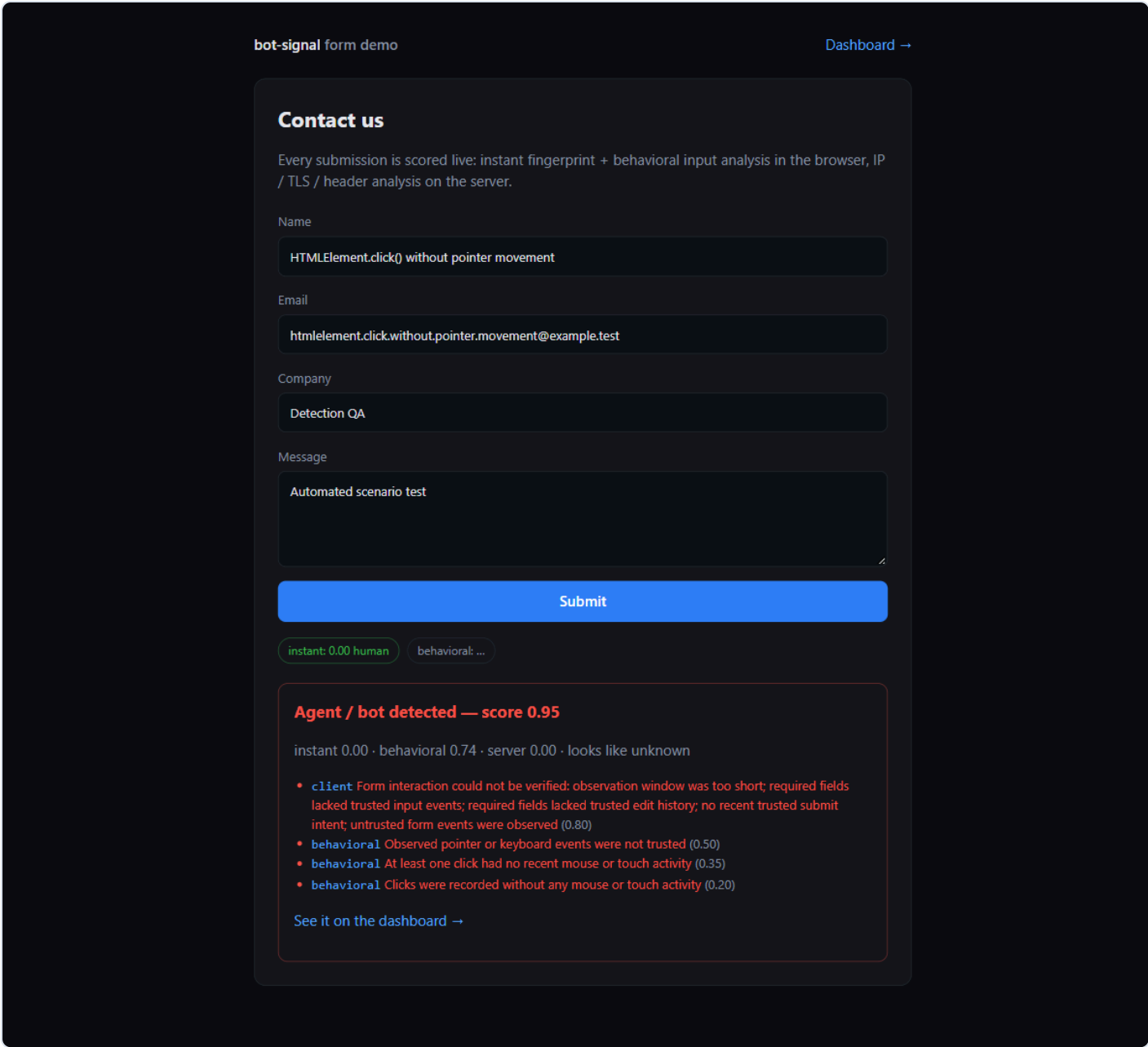
instant 0.00 · behavioral 0.69 · server 0.00 · looks like unknown

behavioral Typing reused a short programmed delay pattern (0.55)

behavioral Every measured click released on the exact pixel it pressed, or too fast to be a finger (0.30)

See it on the dashboard →

Codex/Patchright human-like CDP input: incorrectly classified HUMAN, score 0.00.



JavaScript `element.click()` without pointer movement: correctly classified AGENT, score 0.95.

Contact us

Every submission is scored live: instant fingerprint + behavioral input analysis in the browser, IP / TLS / header analysis on the server.

Name

Headless Patchright

Email

headless@example.test

Company

Detection QA

Message

Automated scenario test

Submit

instant: 1.00 agent

behavioral: ...

Agent / bot detected — score 1.00

instant 1.00 · behavioral 0.00 · server 0.90 · looks like browser-automation

- instant HeadlessChrome user agent/appVersion or webdriver flag (0.90)
- server User-Agent claims a known bot, HTTP, or automation client (0.90)
- client Form interaction could not be verified: observation window was too short (0.80)
- instant User agent is malformed or identifies a known bot, scripting, or automation client (0.70)
- instant WebGL uses a software renderer (SwiftShader/llvmpipe) (0.60)
- instant Notification and Permissions API states contradict (0.55)
- instant Chromium user agent without window.chrome (0.35)
- instant Window has zero outer size or no browser chrome at the screen origin (0.30)
- instant WebGPU shader-f16 feature is missing on Chromium (0.30)
- instant Desktop Chromium with an empty plugin list (0.25)
- instant Network Information reports a zero RTT outside Android (0.20)

See it on the dashboard →

Standard headless automation: correctly classified AGENT, score 1.00.

9. OS-level input injection

The sharpest class of agent is one that drives a real, signed-in browser through the Windows input stack — `SendInput` , `keybd_event` , `mouse_event` , `SetCursorPos` , as used by PowerShell and AutoHotkey harnesses. Those events are genuinely `isTrusted` : the browser really did receive them from the operating system, the window is real, the page load is real, and every provenance check passes. Before this run, such a session was classified HUMAN with score 0.00.

The distinguishing artifact is how the text is pushed in. `SendInput` with `KEYEVENTF_UNICODE` sets `wVk = 0` and carries the character in the scan-code field, so Windows reports `VK_PACKET` and Chromium has no physical key to name. Measured directly against Chrome 140 on Windows 11:

INPUT METHOD	KEY	CODE	KEYCODE	ISTRUSTED
<code>SendInput</code> + <code>KEYEVENTF_UNICODE</code>	K	<i>(empty)</i>	231 (<code>VK_PACKET</code>)	true
Scan-code key press (hardware path)	b	KeyB	66	true

The `injected-key-input` signal fires when at least five printable keystrokes, and at least 60% of them, arrive without a physical key behind them — a ratio, so that an emoji picker or a dictated word inside an otherwise hand-typed form is not flagged. The demo server enforces the same rule on the submitted interaction record, so a tampered page bundle cannot simply drop the evidence.

bot-signal form demo

Dashboard →

Contact us

Every submission is scored live: instant fingerprint + behavioral input analysis in the browser, IP / TLS / header analysis on the server.

Name

Kavin Kumar

Email

kavin.real.injection@example.test

Company

Acme Inc.

Message

Following up on the bot-signal demo - could you share pricing?

Submit

instant: 0.00 human

behavioral: 0.70 agent

Agent / bot detected — score 0.94

instant 0.00 · behavioral 0.70 · server 0.00 · looks like unknown

- client Form interaction could not be verified: form text was injected at the OS level (106/106 keystrokes had no physical key) (0.80)
- behavioral Text was injected at the OS level instead of typed on a keyboard (0.70)

See it on the dashboard →

Real PowerShell `SendInput` session — curved `SetCursorPos` pointer paths, randomised per-character delays, `mouse_event` clicks, genuine page load: now classified AGENT, score 0.94, HTTP 403, on `behavioral:injected-key-input` and `client:untrusted-form-interaction` .

A rewrite that injects real virtual-key/scan-code pairs instead of Unicode packets defeats this specific signal — the keystrokes become indistinguishable from hardware. The supporting `zero-jitter-clicks` heuristic covers part of the pointer side, but this class of attack is not closed by client-side signals alone.

10. Independent red-team results

A separate red-team pass built ten bypass methods against this build and read every verdict back from `GET /api/submissions` rather than the screen. Six passed as human. The findings below are reproduced as stated, with what was changed in response.

FINDING	STATUS	RESPONSE
Client scores are advisory. The server accepted any self-consistent <code>{score: 0, signals: []}</code> , so a full page-session replay (no browser at all) and a <code>Runtime.evaluate</code> forge from inside a real page both scored human 0.00.	Fixed	The page now submits the raw sample streams behind its verdict, and the server recomputes the behavioral result itself — the reported score no longer decides anything. Three new gates: <code>missing-behavioral-samples</code> , <code>forged-client-verdict</code> (reported score or signal list disagrees with the samples it was derived from) and <code>impossible-observation-window</code> (claimed observation longer than the page session has existed — a duration comparison, so a visitor with a wrong clock is unaffected). Verified against the red team's own scripts: both now score agent 0.99.
The CDP console probe never fires for an attached <code>Input</code> -domain client, even with <code>Runtime.enable</code> .	Confirmed, open	Reproduced. Three candidate discriminators were measured and <i>all three rejected</i> : (1) CDP input skipping Chromium's rAF-aligned pipeline — it does not, 16.9 ms median gap versus 16.8 ms for real OS input, zero sub-frame gaps in either; (2) fewer coalesced raw events than a polled mouse — 1.01 per <code>pointermove</code> for both injected paths, and any cadence rule is trivially matched once known; (3) CDP driving an unfocused window — <code>document.hasFocus()</code> stays <code>true</code> . No discriminator was shipped, because none survived measurement.
Injected-key-input is effectively a scan-code check — it catches <code>KEYEVENTF_UNICODE</code> and stock <code>pyautogui</code> , but not <code>KEYEVENTF_SCANCODE</code> , CDP <code>Input</code> , or USB-HID.	Accurate, open	Correct as stated. Injection that supplies a real scan code is indistinguishable from hardware at the DOM level; there is nothing left in the key event to compare.
zero-jitter-clicks skipped presses held under 40 ms , so a fast synthetic clicker slipped past regardless of jitter.	Fixed	A press now reads as machine-made if it releases on the pixel it pressed <i>or</i> is gone before a finger could lift. Skipping the quick ones only told a clicker to hurry.

What remains undetectable. Four of the six passing methods are still classified HUMAN, and this is architectural rather than an unfinished heuristic: OS-level injection with real scan codes (PowerShell `SendInput`, patched `pyautogui`) and a raw CDP `Input` client driving a normal Chrome build both produce genuinely trusted events, a real page load and human-grade input. The samples the server now recomputes are honest samples of real input. There is no client-side signal left to add.

Volume and repetition controls

What those vectors cannot hide is repetition — a hand fills this form once, a loop fills it all afternoon. The server now tracks that as a second axis, kept deliberately out of the verdict: `verdict` remains a statement about one submission, `outcome` is what to do about it. A clean-looking submission from an address that has already sent the same content, or too many submissions, or opened too many page sessions, returns `outcome: "review"` and HTTP 202 instead of being silently accepted. Limits are environment-tunable (`MAX_SUBMISSIONS_PER_SESSION`, `MAX_SUBMISSIONS_PER_IP`, `MAX_SESSIONS_PER_IP`, `MAX_CHALLENGES_PER_SESSION`, `MAX_IDENTICAL_SUBMISSIONS`, `RISK_WINDOW_MS`).

Verified against the scan-code injection that defeats detection outright: the first run scored `human / accepted`, and the second run with the same content scored `human / review`. The verdict stayed honest; the submission stopped being accepted in silence. This is mitigation, not detection, and is reported as such.

A bug this suite could not see

Routing the recomputed behavioral layer through the existing client-layer validation made `invalid-client-layer` fire on every samples-backed submission, including honest ones — the validator requires `triggered === true` on each signal, and a full analysis carries untriggered entries too. All 81 tests stayed green, because every case in this matrix expects AGENT, so a verdict that was right for the wrong reason was indistinguishable from one that was right. It surfaced only when a real scan-code injection run came back AGENT on a signal it had no business tripping. A false-positive guard now asserts that a genuine page session carrying real samples trips none of the four sample-integrity gates.

11. Scope note

This is a risk-based set of 66 classification scenarios covering the app's documented layers and the requested Chrome/DOM/JavaScript/request techniques. "All possible" browser and network combinations are not finite; the report does not certify every OS, browser version, mobile device, assistive technology, extension, proxy or TLS terminator.

Chrome DevTools MCP was used directly for native MCP form filling/clicking and MCP-executed `HTMLInputElement.click()`, `form.requestSubmit()`, and synthetic submit dispatch. The wider matrix additionally used installed Google Chrome through Patchright and raw requests.

16 additional functionality, false-positive-guard and security checks also passed (page/dashboard loading, validation, HTTP handling, storage limits, Unicode, dashboard consistency/reset and output encoding). They are excluded from the headline numbers because they do not answer HUMAN versus AGENT.